Tutorial - 1

Q 1: Describe all the 2x2 row-reduced echelon matrices

Sol^n Let A be a 2x2 matrix in row reduced echelon form.
~~(All the entries of A will either be 0 or 1)~~
then, A can have 2 pivots or 1 pivot or no pivots at all
Thus, A can be

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & b \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad b \in \mathbb{C}.$$

Q 2: Let $A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 0 \end{bmatrix}$ For which $y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$ does the System $AX = y$ have a solution?

Sol^n Let $A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 0 \end{bmatrix}$; $y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in R^3$

then the augmented matrix of system $AX = y$ is

$$\begin{bmatrix} 3 & -1 & 2 & ; & y_1 \\ 2 & 1 & 1 & ; & y_2 \\ 1 & -3 & 0 & ; & y_3 \end{bmatrix}$$

$R_2 \to R_2 - \frac{2}{3}R_1$, $R_3 \to$ ~~$R_3 \to R$~~ $R_3 - \frac{1}{3}R_1$

$$\begin{bmatrix} 3 & -1 & 2 & y_1 \\ 0 & 5/3 & -1/3 & y_2 - \frac{2}{3}y_1 \\ 0 & -8/3 & -2/3 & y_3 - \frac{y_1}{3} \end{bmatrix} \quad R_3 \to R_3 + \frac{8}{5}R_2 \quad \begin{bmatrix} 3 & -1 & 2 & y_1 \\ 0 & \frac{5}{3} & -1/3 & y_2 - \frac{2}{3}y_1 \\ 0 & 0 & -\frac{6}{5} & -\frac{7}{3}y_1 + \frac{8}{5}y_2 + y_3 \end{bmatrix}$$

Now, write the corresponding system of equation

$$3x_1 - x_2 + 2x_3 = y_1 \quad \text{--- (1)}$$

$$\frac{5}{3}x_2 - \frac{1}{3}x_3 = y_2 - \frac{2}{3}y_1 \quad \text{--- (2)}$$

$$-\frac{6}{5}x_3 = -\frac{7}{3}y_1 + \frac{8}{5}y_2 + y_3 \quad \text{--- (3)}$$

$$\Rightarrow x_3 = \frac{35}{18}y_1 - \frac{4}{3}y_2 - \frac{5}{6}y_3$$

putting $x_3 = \frac{35}{18}y_1 - \frac{4}{3}y_2 - \frac{5}{6}y_3$ in (2) we can get x_2

Similarly, by putting the values of x_3, x_2 in (1) we can get x_1

therefore, the system $AX = Y$ has a unique sol. for any $y \in R^3$

i.e. then $AX = Y$ has a unique sol $\forall\ Y \in \mathbb{R}^3$

Q3: Let A be a nxn matrix

(i) Suppose there exists a nxn matrix B such that $BA = I$. Show that A is invertible.

Sol^n suppose $\exists$ a $_{n\times n}$ matrix $B_{n\times n}$ s.t. $BA = I$

consider the system $AX = 0$

clearly, $X = 0$ is a solution of the system $AX = 0$

Consider, $AX = 0$

multiply on left by B

$B(AX) = B0 = 0$

$\Rightarrow (BA)X = 0$

$\Rightarrow IX = 0$

$\Rightarrow X = 0$

Thus, $X = 0$ is the only sol. of the system $AX = 0$

Since, A is invertible iff $AX = 0$ has a unique sol.

$\therefore$ A is invertible

(ii) Suppose there exists a nxn matrix C such that $AC = I$. Show that 'A' is invertible.

Sol^n Suppose $\exists$ a nxn matrix C such that $AC = I$

consider the system $CX = 0$

clearly, $X = 0$ is the sol. of the system $CX = 0$

consider, $CX = 0$

multiply on left by A

$A(CX) = A0$

$(AC)X = A0 = 0$

$\Rightarrow IX = 0$

$\Rightarrow X = 0$

Thus, $X = 0$, is the only solution of the system $CX = 0$

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Since, any matrix M is invertible iff $MX=0$ has a unique sol.

$\therefore$ C is Invertible

By definition, $\exists D \in M_{n\times n}(F)$ s.t. $CD = DC = I$

Now, $ID = (AC)D = A(CD) = A(I) = A$

$\therefore$ $AC = CA = I$

$\Rightarrow$ A is invertible and $A^{-1} = C$

Q4: Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -4 & 4 \end{bmatrix}$

Find a matrix C such that $CA = B$. Is the matrix C unique?

Sol^n $A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -4 & 4 \end{bmatrix}$

We need to find matrix C such that $CA = B$.

The order of C must be 2×3

Let $C = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \end{bmatrix}$ s.t. $CA = B$

$$\Rightarrow \begin{pmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -4 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} c_1+2c_2+c_3 & -c_1+2c_2 \\ c_4+2c_5+c_6 & -c_4+2c_5 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -4 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{aligned} c_1+2c_2+c_3 &= 3 \\ -c_1+2c_2 &= 1 \\ c_4+2c_5+c_6 &= -4 \\ -c_4+2c_5 &= 4 \end{aligned}$$

Solve the system using Gaussian elimination.

$$\left[\begin{array}{cccccc|c} 1 & 2 & 1 & 0 & 0 & 0 & 3 \\ -1 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 & -4 \\ 0 & 0 & 0 & -1 & 2 & 0 & 4 \end{array}\right]$$

$R_1 \to R_2 + R_1$, $R_4 \to R_4 + R_3$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 1 & 0 & 0 & 0 & 3 \\ 0 & 4 & 1 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 & 1 & -4 \\ 0 & 0 & 0 & 0 & 4 & 1 & 0 \end{array}\right]$$

$R_2 \to \frac{1}{4} R_2$ & $R_4 \to \frac{1}{4} R_4$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 1/4 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 & -4 \\ 0 & 0 & 0 & 0 & 1 & 1/4 & 0 \end{array}\right]$$

$R_1 \to R_1 - 2R_2$ & $R_3 \to R_3 - 2R_4$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 1/2 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/4 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1/2 & -4 \\ 0 & 0 & 0 & 0 & 1 & 1/4 & 0 \end{array}\right]$$

Convert back to system of equations

$$c_1 + \frac{c_3}{2} = 1 \Rightarrow c_1 = 1 - c_3/2$$

$$c_2 + c_3/4 = 1 \Rightarrow c_2 = 1 - c_3/4$$

$$c_4 + \frac{c_6}{2} = -4 \Rightarrow c_4 = -4 - c_6/2$$

$$c_5 + \frac{c_6}{4} = 0 \Rightarrow c_5 = -\frac{c_6}{4}, \quad c_3, c_6 \in \mathbb{R}$$

$$\therefore \quad C = \begin{bmatrix} 1 - c_3/2 & 1 - c_3/4 & c_3 \\ -4 - \frac{c_6}{2} & -\frac{c_6}{4} & c_6 \end{bmatrix}, \quad c_3, c_6 \in \mathbb{R}$$

Take $c_3 = 4\alpha$, $c_6 = 4\beta$ for $\alpha, \beta \in \mathbb{R}$, then

$$C = \begin{bmatrix} 1-2\alpha & 1-\alpha & 4\alpha \\ -4-2\beta & -\beta & 4\beta \end{bmatrix} \quad \alpha, \beta \in \mathbb{R}$$

In particular, take $\alpha = \beta = 0$, then $C = \begin{bmatrix} 1 & 1 & 0 \\ -4 & 0 & 0 \end{bmatrix}$

As α, β can take any real value. So such C is not indeed there are infinitely many such matrix C.

Q5: If A is an $m\times n$ matrix and B is an $n\times m$ matrix with $n<m$, show that AB is not invertible.

Sol^n: $A_{m\times n}$, $B_{n\times m}$ $(n<m)$

Claim: $\exists\, x_0 \neq 0$ such that $ABx_0 = 0$

Since, $n<m$ $\exists\, x_0 \neq 0$ s.t. $Bx_0 = 0$

This implies $A(Bx_0) = A\cdot 0 = 0$

$\Rightarrow ABx_0 = 0$ ☑

Q6: Let $B \in M_{n\times n}(\mathbb{C})$

(i) Show that $B' = \{A \in M_{n\times n}(\mathbb{C}) : AB = BA\}$ is a subspace of $M_{n\times n}(\mathbb{C})$.

(ii) Find B', where $B = \begin{bmatrix} 1 & & & 0 \\ & 2 & & \\ & & 3 & \\ 0 & & \ddots & n \end{bmatrix}_{n\times n}$

Sol^n (i) Define, $B' = \{A \in M_{n\times n}(\mathbb{C}) : AB = BA\}$

To show: B' is a subspace of $M_{n\times n}(\mathbb{C})$

Let $A_1, A_2 \in B'$ then $A_1B = BA_1$ and $A_2B = BA_2$ and $\alpha \in \mathbb{C}$

Consider $(\alpha A_1 + A_2)B = (\alpha A_1)B + A_2B$

$= \alpha(A_1B) + A_2B$

$= \alpha(BA_1) + (BA_2)$ $(\because A_1B = BA_1 \,\&\, A_2B = BA_2)$

$= B(\alpha A_1 + A_2)$

$\Rightarrow \alpha A_1 + A_2 \in B' \ \forall \alpha \in \mathbb{C}\ \forall A_1, A_2 \in B'$

$\therefore$ B' is a subspace of $M_{n\times n}(\mathbb{C})$ (By subspace test)

(ii) $B' = \{A \in M_{n\times n}(\mathbb{C}) : AB = BA$, $B = \begin{bmatrix} 1 & & & 0 \\ & 2 & & \\ & & 3 & \\ 0 & & \ddots & n \end{bmatrix}$

Let $A \in B'$ then $AB = BA$

Let B_{ij} be the element of i^{th} row and j^{th} column of matrix B.

Then $B_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ n & \text{if } i = j = n \end{cases}$

observe that, $(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$

As $B_{ij} = 0$ if $i \neq j$ and $B_{ij} = n$ if $i = j = n$

So $(AB)_{ij} = A_{ij} B_{jj} = j A_{ij}$

Similarly $(BA)_{ij} = \sum_{k=1}^{n} B_{ik} A_{kj} = B_{ii} A_{ij} = i A_{ij}$

As $AB = BA$

So, $(AB)_{ij} = (BA)_{ij}$

$\Rightarrow \quad j A_{ij} = i A_{ij}$

$\Rightarrow \quad (j - i) A_{ij} = 0$ —— $(*)$

If $i = j$ (diagonal entries) then $(*)$ holds true for any A_{ij}

if $i \neq j$ then we must have $A_{ij} = 0$

Thus, $A_{ij} = 0$ whenever $i \neq j \Rightarrow A$ is any diagonal matrix in $M_{n\times n}(\mathbb{C})$

$\therefore \quad B' = \{ A \in M_{n\times n}(\mathbb{C}) : A_{ij} = 0 \text{ whenever } i \neq j \}$

$= \left\{ \begin{bmatrix} \alpha_1 & \cdots & & 0 \\ \vdots & \alpha_2 & \cdots & 0 \\ & & \alpha_3 & \vdots \\ 0 & - & - & \alpha_n \end{bmatrix} : \alpha_i \in \mathbb{C} \ \forall \ 1 \leq i \leq n \right\}$ ☒

<u>Q7:</u> Let $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in M_{n\times 1}(\mathbb{C})$. consider the set

$LA(X) = \{ A \in M_{n\times n}(\mathbb{C})$ ~~is a subspace of $M_{n\times n}(\mathbb{C})$~~ $: AX = 0 \}$

(i) Show that $LA(X)$ is a subspace of $M_{n\times n}(\mathbb{C})$

(ii) Show that $LA(X)$ is a left ideal of $M_{n\times n}(\mathbb{C})$, that is $BA \in LA(X)$ whenever $A \in LA(X)$ and $B \in M_{n\times n}(\mathbb{C})$

(iii) Show that $LA(X) = M_{n\times n}(\mathbb{C})$ iff $X = 0$

Sol^n: (i) Let $A, B \in LA(X) \Rightarrow AX=0$ and $BX=0$

$\alpha \in \mathbb{C}$

Consider, $(\alpha A+B)X$

$= \alpha(AX)+BX$

$= \alpha \cdot 0 + 0$ $\quad (\because AX=0=BX)$

$\therefore \alpha A+B \in LA(X) \ \forall \alpha \in \mathbb{C} \ \forall A, B \in LA(X)$

$\therefore LA(X)$ is a subspace of $M_{n\times n}(\mathbb{C})$

(ii) Let $B \in M_{n\times n}(\mathbb{C})$, $A \in LA(X)$

Clearly, the matrix product BA is well defined

As $A \in LA(X) \Rightarrow AX=0$

Consider, $(BA)X$

$= B(AX)$

$\Rightarrow B \cdot 0 = 0$

$BA \in LA(X)$

$\therefore LA(X)$ is a left ideal of $M_{n\times n}(\mathbb{C})$

(iii) Suppose that $LA(X) = M_{n\times n}$

Let $A \in M_{n\times n}$ then $A \in M_{n\times n} = LA(X) \Rightarrow AX=0$

i.e. $AX=0 \ \forall A \in M_{n\times n}$

In particular for $A=I$, we get

$AX = IX = 0 \Rightarrow X=0$

Conversely, let $X=0$, and $A \in M_{n\times n}(\mathbb{C})$.

then, $AX = A0 = 0 \Rightarrow A \in LA(X)$

$M_{n\times n}(\mathbb{C}) \subseteq LA(X)$

Also, $LA(X) \subseteq M_{n\times n}(\mathbb{C})$

$\therefore M_{n\times n}(\mathbb{C}) = LA(X)$

Q8: Let $A, B \in M_{n\times n}(\mathbb{C})$ can be such that $AB=0$. Give a proof or counter example for each of the following.

(i) $BA=0$

(ii) Either $A=0$ or $B=0$

(iii) If B is invertible then $A=0$

(iv) There is a vector $X \neq 0$ such that $BAX = 0$

Sol^n: (i) False, Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$

then $AB = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

and $BA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$

(ii) False, Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ then $AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

(iii) True, Let $A, B \in M_{n\times n}(\mathbb{C})$ s.t. $AB = 0$ and B is invertible

As B is invertible

$\exists$ $C \in M_{n\times n}(\mathbb{C})$ s.t. $BC = CB = I$

Consider, $AB = 0$

Multiply both side by C (on right)

$\Rightarrow (AB)C = 0C$

$\Rightarrow A(BC) = 0$

$\Rightarrow AI = 0$

$\Rightarrow A = 0$

(iv) True, Let If possible $\nexists$ any non-zero $X \in M_{n\times 1}(\mathbb{C})$ s.t. $BAX = 0$ i.e. the system $BAX = 0$ has a unique sol. namely $X = 0$. As any matrix M is invertible iff $MX = 0$ has a unique sol. $\therefore$ BA is invertible

$\exists$ $C \in M_{n\times n}(\mathbb{C})$ such that $(BA)C = C(BA) = I$, I denotes $n\times n$ Identity matrix

Now, $C(BA) = I$

Multiply both side by B (on right)

$\Rightarrow$ $C(BA)B = IB = B$

$\Rightarrow$ $CB(AB) = B$

$\Rightarrow$ $CB(0) = B$ $\Rightarrow B = 0$ $(\because AB = 0)$

$\therefore$ ~~AB~~ $BA = 0A = 0$ which is a contradiction to the fact that BA is invertible ($\because$ Zero matrix is not invertible.

Tutorial-2

Q 1: Are the vectors $a_1=(1,1,2,4)$, $a_2=(2,-1,-5,2)$, $a_3=(1,-1,-4,0)$, $a_4=(2,1,1,6)$ linearly dependent in $\mathbb{R}^4$?

Sol^n $a_1=(1,1,2,4)$, $a_2=(2,-1,-5,2)$, $a_3=(1,-1,-4,0)$
$a_4=(2,1,1,6)$.

Let $\sum_{i=1}^{4} \alpha_i a_i = 0 = (0,0,0,0)$

$\Rightarrow \alpha_1(1,1,2,4)+\alpha_2(2,-1,-5,2)+\alpha_3(1,-1,-4,0)+\alpha_4(2,1,1,6)=0$

$\Rightarrow (\alpha_1+2\alpha_2+\alpha_3+2\alpha_4,\ \alpha_1-\alpha_2-\alpha_3+\alpha_4,\ 2\alpha_1-5\alpha_2-4\alpha_3+\alpha_4,\ 4\alpha_1+2\alpha_2+0\alpha_3+6\alpha_4) = 0 = (0,0,0,0)$

$$\Rightarrow \left.\begin{aligned} \alpha_1+2\alpha_2+\alpha_3+2\alpha_4&=0 \\ \alpha_1-\alpha_2-\alpha_3+\alpha_4&=0 \\ 2\alpha_1-5\alpha_2-4\alpha_3+\alpha_4&=0 \\ 4\alpha_1+2\alpha_2+0\alpha_3+6\alpha_4&=0 \end{aligned}\right\} - (*)$$

Use Gaussian elimination method to solve for α_i's

$$\left[\begin{array}{cccc:c} 1 & 2 & 1 & 2 & 0 \\ 1 & -1 & -1 & 1 & 0 \\ 2 & -5 & -4 & 1 & 0 \\ 4 & 2 & 0 & 6 & 0 \end{array}\right]$$

$R_2 \to R_2-R_1,\ R_3 \to R_3-2R_1,\ R_4 \to R_4-4R_1$

$$\left[\begin{array}{cccc:c} 1 & 2 & 1 & 2 & 0 \\ 0 & -3 & -2 & -1 & 0 \\ 0 & -9 & -6 & -3 & 0 \\ 0 & -6 & -4 & -2 & 0 \end{array}\right]$$

$R_3 \to R_3+3R_2$ & $R_4 \to R_4+2R_2$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & -3 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Convert back to eqn

$$\alpha_1 + 2\alpha_2 + \alpha_3 + 2\alpha_4 = 0 \quad \text{---(1)}$$
$$-3\alpha_2 - 2\alpha_3 - \alpha_4 = 0 \quad \text{---(2)}$$

$$\alpha_2 = -\frac{2}{3}\alpha_3 - \frac{1}{3}\alpha_4$$

put $\alpha_2 = -\frac{2}{3}\alpha_3 - \frac{1}{3}\alpha_4$ in (1)

$$\Rightarrow \alpha_1 + 2\left(-\frac{2}{3}\alpha_3 - \frac{\alpha_4}{3}\right) + \alpha_3 + 2\alpha_4 = 0$$

$$\Rightarrow \alpha_1 = \frac{1}{3}\alpha_3 - \frac{4}{3}\alpha_4$$

Thus, $\alpha_1 = \frac{1}{3}\alpha_3 - \frac{4}{3}\alpha_4$, $\alpha_2 = -\frac{2}{3}\alpha_3 - \frac{1}{3}\alpha_4$, $\alpha_3, \alpha_4 \in \mathbb{R}$

(i.e. α_3, α_4 are free to take any real value)

The system of equation - (*) has infinitely many sol.

i.e. $\exists$ α_i's not all zero s.t. $\alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3 + \alpha_4 a_4 = 0$

$\therefore$ The vectors a_1, a_2, a_3, a_4 are L.D. in $\mathbb{R}^4$.

Q2: Let V be the v.s. of 2×2 matrices over $\mathbb{C}$

(i) prove that V has dim 4 by exhibiting a basis for V which has four elements.

Sol^n $V = M_{2\times2}(\mathbb{C})$

(i) Consider, $\beta = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \subseteq M_{2\times2}(\mathbb{C}) = V$

Claim: β is a basis for V.

• **β is L.I**

Let $\alpha_i \in \mathbb{C}$ s.t. $\sum_{i=1}^{4} \alpha_i v_i = 0$

i.e. $\alpha_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \alpha_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} \alpha_1 & \alpha_2 \\ \alpha_3 & \alpha_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$

$\therefore \beta = \{v_1, v_2, v_3, v_4\}$ is L.I. set.

• **β is a spanning set**

Let $A \in M_{2\times 2}(\mathbb{C}) \Rightarrow A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ for $a_{ij} \in \mathbb{C}$

Observe,

$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_{12} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + a_{13} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$
$+ a_{14} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \in span(\beta)$.

$\therefore V \subseteq Span(\beta)$

Also $Span(\beta) \subseteq V$

$\therefore V = span(\beta)$

$\therefore \beta$ is a basis for V.

Thus, $\dim(V) = \# \beta = 4$

(ii) Let $W = \{A \in M_{2\times 2}(\mathbb{C}) : A_{11} + A_{22} = 0\}$. Show that W is a subspace of V and find the dim. of W.

Sol^n $W = \{A \in M_{2\times 2}(\mathbb{C}) : A_{11} + A_{22} = 0\}$

For any matrix $A \in M_{2\times 2}(\mathbb{C})$ whose i^{th} row and j^{th} row entry is A_{ij}, $Trace(A) = A_{11} + A_{22}$.

So we can define W as

$W = \{A \in M_{2\times 2}(\mathbb{C}) : Tr(A) = 0\}$

As $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in W$

clearly W is non-empty

Let $\alpha \in \mathbb{C}$, $A, B \in M_{2\times2}(\mathbb{C})$

$\Rightarrow Tr(A) = Tr(B) = 0$

Consider $Tr(\alpha A + B) = \alpha Tr(A) + Tr(B)$

$= \alpha \cdot 0 + 0$

$= 0$

$\Rightarrow \alpha A + B \in W$

$\therefore$ W is a subspace of $V (= M_{2\times2}(\mathbb{C}))$

Consider, $\mathcal{V} = \left\{ \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$

<u>claim:</u> $\mathcal{V}$ is a basis of W

As $Tr\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = Tr\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = Tr\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\right) = 0$

$\Rightarrow \mathcal{V} \subseteq W$

• <u>$\mathcal{V}$ is L.I</u> Let $\alpha_i \in \mathbb{C}$ s.t. $\alpha_1 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\Rightarrow \begin{bmatrix} \alpha_3 & \alpha_2 \\ \alpha_1 & -\alpha_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$

$\therefore$ <u>$\mathcal{V}$ is L.I.</u>

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- $\mathcal{V}$ is a spanning set for W

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in W \Rightarrow Tr(A) = a_{11} + a_{22} = 0$

i.e. $a_{22} = -a_{11}$

$\therefore A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & -a_{11} \end{bmatrix} = a_{21}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + a_{12}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + a_{11}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in$ span

$\therefore W \subseteq span(V)$

$\therefore W = span(V)$

Thus, $\mathcal{V} = \left\{ \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$ is a basis of W.

$\therefore \dim(W) = \# \mathcal{V} = 3$

Q3: Let X be the vector space of all pol. $p(t)$ of deg. $<n$ and let Y be the set of pol. in X that are zero at $t_1, t_2 \cdots t_j$; $j < n$

Sol^n $X = \{ p(t) \mid p(t) = \sum_{i=1}^{n-1} a_i t^i, \ a_i \in \mathbb{C} \} = P_{n-1}(\mathbb{C})$

For $j < n$, $i \le n-1$

Define $Y = \{ p(t) \in X \mid p(t_i) = 0 \text{ for } i = 1, 2, \cdots j \}$

(i) Show that Y is a subspace of X

Sol^n Clearly, zero pol. is there in W

so, W is not empty.

Let $\alpha \in \mathbb{C}$, $f_1, f_2 \in Y$

$\Rightarrow f_1(t_k) = f_2(t_k) = 0$ for $k = 1, 2, \cdots j$ —(i)

Consider, $(\alpha f_1 + f_2) t_k$

$= \alpha f_1(t_k) + f_2(t_k) = 0$ By (i)

$\therefore (\alpha f_1 + f_2)\, t_k = 0$

$\therefore \alpha f_1 + f_2 \in Y \quad \forall \alpha \in \mathbb{C} \;\; \forall f_1, f_2 \in Y$

hence, Y is a subspace of X.

(ii) Find $\dim X$ and $\dim Y$

Consider the subset $\beta = \{1, t, t^2 \cdots t^{n-1}\}$ of X

- β is L.I. Let $\alpha_i \in \mathbb{C}$ s.t. $\sum_{i=0}^{n-1} \alpha_i t^i = 0$

If any $\alpha_i \neq 0$ then $\sum_{i=0}^{n-1} \alpha_i t^i$ is a non-zero pol. of degree $\leq n-1$ with infinitely many roots, which contradicts the fundamental thm of algebra

$\therefore \alpha_i = 0 \;\forall\; i = 1, 2, \cdots n-1$

- β is a spanning set of X

Let $f(t) \in X$

$\Rightarrow f(t) = a_0 + a_1 t + \cdots + a_{n-1} t^{n-1} \in \operatorname{span}(\beta)$

$\therefore X \subseteq \operatorname{span}(\beta)$

Thus $X = \operatorname{span}(\beta)$

Thus $\beta = \{1, t, t^2 \cdots t^{n-1}\}$ forms a basis for X

$\therefore \dim(X) = \# \beta = n.$

Define, $f(t) = (t - t_1)(t - t_2) \cdots (t - t_j) = \prod_{k=1}^{j} (t - t_k) \quad (j \leq n-1)$

and $\gamma = \{f(t), t f(t), t^2 f(t) \cdots t^{n-1-j} f(t)\}$

observe that, $f(t_k)=0 \ \forall \ k=1,2,3,\dots,j$

$\therefore \ t_k^i f(t_k)=0 \ \forall \ i=0,1,\dots,n-1-j$ and $k=1,2,\dots,j$

$\therefore \ \mathcal{V} \subseteq Y$

- $\underline{\mathcal{V} \text{ is L.I}}$

Let $d_i \in \mathbb{C}$ s.t. $\sum_{i=0}^{n-1-j} d_i t^i f(t)=0$

If any of d_i is non-zero then it contradicts the ~~fact that~~ f.T.A. $\therefore \ d_i=0 \ \forall \ i=0,1,2,\dots,n-1-j$

- $\underline{\mathcal{V} \text{ is a spanning set of } Y}$

Let $p(t) \in Y \Leftrightarrow p(t_k)=0$ for $k=1,2,\dots,j$

$\Leftrightarrow (t-t_k)$ is a factor of $p(t)$ for $k=1,\dots,j$

$\Leftrightarrow p(t)=p_1(t)(t-t_1)(t-t_2)\dots(t-t_j)$

~~which~~ with $\deg(p_1(t)) \leq n-1-j$

i.e. $p(t)=p_1(t)f(t)$

where $p_1(t)=\sum_{k=0}^{n-1-j} c_k t^k$

$$\Leftrightarrow p(t)=\sum_{k=0}^{n-1-j} c_k t^k f(t) \in \operatorname{Span}(\mathcal{V})$$

$\therefore \ Y=\operatorname{span}(\mathcal{V})$

Thus, $\mathcal{V}$ forms a basis for Y

so, $\dim(Y)=\#\mathcal{V}=n-j$

(iii) $\dim(X/Y)=\dim X-\dim(Y)=(n)-(n-j)$

$=j$

Q 4: Let V be a V.S. and 'A' is L.I. subset of V. Prove that 'A' is a basis for V iff it is a maximal linearly independent subset of V. i.e. addition of any vector to A will results in a L.D. set.

Sol^n Suppose that A is a L.I. subset of V.S. V over F.

To prove: 'A' is a basis iff A is a maximal linearly independent subset of V.

$\Rightarrow$ Suppose that A is a basis for V.

Then, A is L.I. and $V = \text{Span}(A)$

Let $v \in V$ s.t. $v \notin A$

It is enough to show that $\{v\} \cup A$ is L.D.

As $v \in V = \text{Span}(A)$

$v = \sum_{i=1}^{n} \alpha_i v_i$ where $\alpha_i \in F$ & $v_i = A$ for all $1 \le i \le n$.

$\Rightarrow 1 \cdot v + (-\alpha_1) v_1 + \cdots + (-\alpha_n) v_n = 0$

$\Rightarrow \{v, v_1, \cdots v_n\}$ is dependent.

$\Rightarrow \{v\} \cup A$ is dependent.

($\because$ Every superset of a linearly dependent set is L.D.)

Thus, A is maximal linearly independent set of V.

$\Leftarrow$ Let A be the maximal linearly independent subset of V.

It is enough to show $V = \text{span}(A)$

Let $v \in V$

if $v \in A$ then $v \in span(A)$

if $v \notin A$ then $\{v\} \cup A$ is L.D.

($\because$ A is the maximal linearly independent subset)

So $\exists$ δ_i's not all zero s.t.

$$\delta_1 v + \sum_{i=2}^{m} \delta_i v_i = 0 \text{ where } v_i \in A \text{ for } 2 \le i \le m$$

Claim $\delta_1 \neq 0$

If $\delta_1 = 0$ then $\sum_{i=2}^{m} \delta_i v_i = 0$ but not all δ_i's are zero

$\Rightarrow$ $\{v_2, v_3 \dots v_m\}$ is L.D. which is a contradiction

as $\{v_2, v_3 \dots v_m\} \subseteq A$ and every subset of L.I. set is L.I.

$\therefore$ $\delta_1 \neq 0$

$\therefore$ $v = \sum_{i=2}^{m} (-\delta_i \delta_1^{-1}) v_i \in Span(A)$

$\therefore$ $V = Span(A)$

Qs: Let V be a vector space, and let $span(A) = V$, for a subset of V. Prove that A is a basis for V iff it is a minimal spanning set, that is removal of any vector from A will result in a set which does not span V.

Sol^n Let V be a vector space and let $span(A) = V$ for a subset A of V.

To prove: A is a basis of V iff it is a minimal spanning set.

Suppose that A is a basis of V.

$\Rightarrow$ 'A' is L.I set and $span(A) = V$

we need to show, A is minimal spanning set for V.

Let $v \in A$, define $B = A \setminus \{v\}$ then $A = B \cup \{v\}$

Claim: $v \notin span(B)$

Let if possible, $v \in span(B)$

$\Rightarrow \quad v = \sum_{i=1}^{m} \alpha_i v_i \quad$ for $\alpha_i \in F$, $v_i \in B$

As $v_i \in B$ & $v_i \neq v \ \forall i$

$= A \setminus \{v\}$

$\Rightarrow \quad 1 \cdot v + (-\alpha_1) v_1 + (-\alpha_2) \cdot v_2 + \cdots + (-\alpha_m) v_m = 0$

$\Rightarrow \{v, v_1, \cdots v_m\}$ is L.D. set.

$\Rightarrow \{v, v_1, v_2, \cdots, v_m\} \subseteq A$ and every subset of a l.D set is l.D.

So, A is a l.D. set which is a contradiction to the fact that A is a basis.

$\therefore$ claim holds and $v \notin span(B)$

$\therefore \quad span(B) \neq V \quad (\because v \in V$ but $v \notin span(B))$.

(Thus if we remove any vector from set A, the remaining vectors can't form a spanning set for V. So A is minimal spanning set.)

$\Leftarrow$ Conversely, suppose that A is the minimal spanning set. In order to prove that A is a basis for V, it is enough to show that A is L.I.

Let $\alpha_i \in F$ s.t. $\sum_{i=1}^{k} \alpha_i v_i = 0$ where $v_i \in A$ for $i=1,2,\dots,k$

If any of the α_i's in non-zero then WLOG we can assume that α_1 is non-zero

Then $$\alpha_1 v_1 + \sum_{i=2}^{k} \alpha_i v_i = 0$$

$$\Rightarrow v_1 = \sum_{i=2}^{k} (-\alpha_1^{-1}\alpha_i) v_i$$

So $\text{span}(A) = \text{span}(A \setminus \{v_1\}) = V$

$\because A \setminus \{v_1\} \subseteq A \Rightarrow \text{Span}(A \setminus \{v_1\}) \subset \text{span}(A)$

Let $x \in \text{Span} A \Rightarrow x = \sum_{i=1}^{r} \alpha_i' y_i \quad y_i \in A \ \forall i \ \& \ \alpha_i' \in F \ \forall i$

if $y_i \neq v_1$ for $1 \leq i \leq r$ then $y_i \in A \setminus \{v_1\}$

$\therefore x \in \text{Span}(A \setminus \{v_1\})$

If $y_{i_0} = v_1$ for some i_0 then put $y_{i_0} = v_1 = \sum_{i=2}^{k} (-\alpha_1^{-1}\alpha_i) v_i$

then
$$x = \sum_{\substack{i=1 \\ i \neq i_0}}^{r} \alpha_i' y_i + \sum_{i=2}^{k} (-\alpha_1^{-1}\alpha_i) v_i \in \text{Span}(A \setminus \{v_1\})$$

Thus, $A \setminus \{v_1\}$ is also a spanning set, which is proper subset of A. this contradicts the fact that A is the minimal spanning set.

Thus, $\alpha_i = 0 \ \forall i$

hence, A is L.I.

Thus A forms a basis for V.

Tutorial-3

Q1: Let $C([0,1]) = \{f : [0,1] \to \mathbb{C} : f \text{ is cts}\}$.
Define $T : C([0,1]) \to \mathbb{C}$ by
$$T(f) = \int_0^1 f(x)x^2\,dx \quad \forall\, f \in C([0,1])$$
Show that T is linear map.

Sol^n $C([0,1]) = \{f : [0,1] \to \mathbb{C} : f \text{ is cts}\}$.
Define $T : C([0,1]) \to \mathbb{C}$ by
$$T(f) = \int_0^1 f(x)x^2\,dx \quad \forall\, f \in C([0,1])$$
Let $\alpha \in \mathbb{C}$, $f, g \in C([0,1])$

$$T(\alpha f+g) = \int_0^1 (\alpha f+g)(x)\,x^2\,dx$$
$$= \int_0^1 (\alpha f)(x)\,x^2\,dx + \int_0^1 g(x)\cdot x^2\,dx$$
$$= \alpha\int_0^1 f(x)\cdot x^3\,dx + \int_0^1 g(x)\cdot x^3\,dx$$
$$= \alpha T(f) + T(g)$$
hence, T is a linear map.

Q2: Let X be a vector space over F. If $x \in X$ is a non-zero vector, show that there exists a linear map $T : X \to F$ s.t. $T(x) \neq 0$

Sol^n: Let X be a vector space over F.
Let $x \in X$ s.t. $x \neq 0$
As every vector space has a basis

There exists a basis β of X.

As $x \in X = \text{Span}(\beta) \Rightarrow x = \sum_{i=1}^{m} \alpha_i v_i$ where $\alpha_i \in F$, $v_i \in \beta$

As $x \neq 0$, all α_i's can't be zero, Atleast one has to be non-zero.

WLOG, assume that α_1 is non-zero.

Now, define a map $T: X \to \mathbb{F}$ as $T\left(\sum_{i=1}^{n} \beta_i v_i\right) = \beta_1$

Claim: T is a linear map such that $T(x) \neq 0$

Let $x_1 = \sum_{i=1}^{n_1} c_i v_i$, $x_2 = \sum_{i=1}^{n_2} d_i v_i \in X$, $\alpha \in F$ then

- $T(x_1+x_2) = T\left(\sum_{i=1}^{n_1} c_i v_i + \sum_{i=1}^{n_2} d_i v_i\right) = T\left((c_1+d_1)v_1 + \sum_{i=2}^{n_1} c_i v_i + \sum_{i=2}^{n_2} d_i v_i\right)$

$= c_1 + d_1$

$= T\left(\sum_{i=1}^{n_1} c_i v_i\right) + T\left(\sum_{i=1}^{n_2} d_i v_i\right)$

$= T(x_1) + T(x_2)$

- $T(\alpha x_1) = T\left(\alpha\left(\sum_{i=1}^{n_1} c_i v_i\right)\right) = T\left(\sum_{i=1}^{n_1} (\alpha c_i) v_i\right) = \alpha c_1 = \alpha T\left(\sum_{i=1}^{n} c_i v_i\right)$

$= \alpha T(x_1)$

hence T is a linear map.

Q3: Let $C([-1,1])$ ~~$\rightarrow C([-1,1])$~~ $= \{ f : [-1,1] \to \mathbb{R} \mid f \text{ is cts} \}$

Define $P: C[-1,1] \to C[-1,1]$ by

$$(Pf)(x) = \frac{f(x)+f(-x)}{2}, \quad \forall x \in [-1,1] \ \& \ f \in C([-1,1]).$$

Show that P is a linear map and P is a Projection that $P^2 = P$.

Sol^n Let $C([-1,1]) = \{f : [-1,1] \to \mathbb{R}, f \text{ is continuous}\}$.

Define $P : C([-1,1]) \to C[-1,1]$ by

$$P(f)(x) = \frac{f(x) + f(-x)}{2} \quad \forall x \in [-1,1] \ \& \ \forall f \in C([-1,1])$$

Let $\alpha \in \mathbb{R}$, $f, g \in C([-1,1])$, then

Now $$P(\alpha f + g)(x) = \frac{(\alpha f + g)(x) + (\alpha f + g)(-x)}{2} \quad \forall x \in [-1,1]$$

$$= \frac{(\alpha f)(x) + g(x) + (\alpha f)(-x) + g(-x)}{2}$$

$$= \frac{\alpha f(x) + g(x) + \alpha f(-x) + g(-x)}{2}$$

$$= \alpha\left(\frac{f(x) + f(-x)}{2}\right) + \left(\frac{g(x) + g(-x)}{2}\right)$$

$$= \alpha P(f) + P(g) \quad \forall f, g \in C([-1,1]).$$

hence P is a linear map.

Consider, $$P^2(f)(x) = P(P(f)(x)) = P\left(\frac{f(x) + f(-x)}{2}\right)$$

$$= \frac{\frac{f(x) + f(-x)}{2} + \frac{f(-x) + f(-(-x))}{2}}{2}$$

$$= \frac{\frac{f(x) + f(-x)}{2} + \frac{f(x) + f(-x)}{2}}{2} = \frac{f(x) + f(-x)}{2}$$

$$= P(f)(-x) \quad \forall x \in [-1,1]$$

Q4: Let X be the vector space of polynomials with complex co-efficients of degree less than n. Let $s_1, s_2, \cdots s_n$ be distinct complex number. Define a map $T: X \to \mathbb{C}^n$ by

$$T(p) = (p(s_1), \cdots, p(s_n)), \ \forall \ p \in X.$$

(i) Show that T is a linear map.

Sol^n Let $p, q \in X$, $\alpha \in \mathbb{C}$

$$
\begin{aligned}
T(\alpha p+q) &= ((\alpha p+q)s_1, (\alpha p+q)s_2, \cdots, (\alpha p+q)s_n) \\
&= ((\alpha p)s_1+q(s_1), (\alpha p)(s_2)+q(s_2), \cdots, (\alpha p)(s_n)+q(s_n)) \\
&= ((\alpha p)s_1, (\alpha p)(s_2), \cdots (\alpha p)s_n) + (q(s_1), q(s_2), \cdots q(s_n)) \\
&= \alpha(p(s_1), p(s_2), \cdots p(s_n)) + (q(s_1), q(s_2), \cdots q(s_n)) \\
&= \alpha T(p) + T(q) \quad \forall \ p, q \in X, \alpha \in \mathbb{C}
\end{aligned}
$$

$\therefore$ T is a linear map.

(ii) The null space of T is trivial and T is not a surjective map.

Sol^n

$$
\begin{aligned}
N_T &= \{p \in X : T(p) = (0, 0, \cdots 0)\} \\
&= \{p \in X : (p(s_1), p(s_2), \cdots p(s_n)) = (0, 0, \cdots 0)\} \\
&= \{p \in X : p(s_i) = 0 \text{ for } i = 1, 2, \cdots n\}
\end{aligned}
$$

If $p \in N_T$ is a constant pol. then p has to be zero polynomial (As $p(s_i) = 0$)

If p ($\neq$ non-constant pol.) $\in N_T$ then p has atleast n roots namely $s_1, s_2, \cdots s_n$ which contradicts the fundamental theorem of Algebra (which says that any non-constant pol. of degree k

k can have atmost k-distinct roots.

here, p can have degree atmost n-1 . hence by FTOA it can have atmost n-1 distinct roots).

$\therefore N_T = \{0\}$

$\therefore \dim N_T = 0$

by dim. theoram, we have

$$\dim N_T + \dim R_T = \dim X$$

$$\Rightarrow \quad 0 + \dim R_T = \dim X = n$$

$$\Rightarrow \quad \dim R_T = n \neq \dim \mathbb{C}^n = n+1$$

As R_T is a Subspace of $\mathbb{C}^n$ and $\dim R_T \neq \dim \mathbb{C}^n = n+1$

$\therefore R_T \subseteq \mathbb{C}^n$

[Note that any $e_j \in \mathbb{C}^{n+1}$ is not there in Range of T]

So T is not a surjective map.

Tutorial- 4

Q1: Let X be a finite dim. vector space. A linear map $M \in L(X,X)$ is similar to $N \in L(X,X)$ if $\exists$ an invertible $S \in L(X,X)$ s.t.

$$M = SNS^{-1}$$

(i) Show that similarity is an equivalence relation.

Sol: $M, N \in L(X,X)$

- **Reflexive** Take $S = I$

  then $M = IMI^{-1} = M$

  M is similar to itself.

- **~~Transitive~~ Symmetric** If M is similar to N

  then, $M = SNS^{-1}$

  pre-multiply by S^{-1} and post multiply by S

  $S^{-1}MS = S^{-1}SNS^{-1}S$ ($\because S$ is invertible)

  $S^{-1}MS = N$

  hence N is similar to M.

- **Transitive:** Let ~~&~~ M is similar to N and N is similar to P

  s.t. $M = SNS^{-1}$

  & $N = TPT^{-1}$ for some invertible matrix T.

then $M = SNS^{-1} = STPT^{-1}S^{-1} = ST\,P(TS)^{-1}$

Since ST is invertible

hence M is similar to P.

(ii) If either M or N in $L(X,X)$ is invertible then MN and NM are similar.

Sol^n case(i) M is invertible

Take $S=M$

Now, $MN = MNMM^{-1}$ $\quad (\because MM^{-1}=I)$
$= M(NM)M^{-1}$

hence MN is similar to NM

case(ii) N is invertible

Take $S=N$

Now, ~~$MN = M$~~ $NM = NMNN^{-1}$ $\quad (\because NN^{-1}=I)$
$= N(MN)N^{-1}$

hence NM is similar to MN.

(iii) If A and B are similar show that $\dim(R_A) = \dim(R_B)$ and $\dim(N_A) = \dim(N_B)$.

Sol^n Let A and B be similar, so $A = SBS^{-1}$ for some invertible S.

Now, $R(A) = \{Ax : x \in X\} = \{SBS^{-1}x : x \in X\}$
$= \{SBy : y \in X\}$ $\quad (y = S^{-1}x)$

Since S is invertible
So $d(R(A)) = d(R(B))$.

Now, $N(A) = \{x \in X : Ax = 0\}$

$= \{x \in X : SBS^{-1}x = 0\}$

$= \{y \in X : SBy = 0\}$ $(y = S^{-1}x)$

Since S is invertible,

$= \{y \in X : SBy = 0$ iff $By = 0$ which is null space of $B\}$

hence $\dim(N(A)) = \dim(N(B))$.

Q4: Let T be a linear map on $\mathbb{R}^2$ defined by

$T(x_1, x_2) = (-x_2, x_1)$

(i) find the matrix representation of T in the standard ordered basis of $\mathbb{R}^2$.

Solⁿ Standard basis of $\mathbb{R}^2 = \{(1,0),(0,1)\} = \beta$

Since T is a linear map.

$T(1,0) = (0,1)$

$T(0,1) = (-1,0)$

$$[T]_\beta = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(ii) Find the matrix representation of T in the ~~standard~~ ordered basis of $\mathbb{R}^2$ $\{(1,2),(1,-1)\}$.

Solⁿ Let $v_1 = (1,2)$

$v_2 = (1,-1)$

$$T(v_1) = T(1,2) = (-2,1) = a(1,2) + b(1,-1)$$
$$T(v_2) = T(1,-1) = (1,1) = c(1,2) + d(1,-1)$$

Further simplifying we get $a = -\frac{1}{3}$, $b = -\frac{5}{3}$
$c = \frac{2}{3}$, $d = \frac{1}{3}$

Thus the matrix rep. is $[T] = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ -\frac{5}{3} & \frac{1}{3} \end{bmatrix}$ //

Q2: Let X be a vector space over F with basis $x = \{x_1, x_2, \cdots, x_n\}$. For each $1 \le i \le n$. Define a linear functional $l_i : X \to F$ by taking linear extensions of the following:

$$l_i(x_j) = \delta_{ij}, \ \forall\ j = 1, 2, \cdots n$$

where $\delta_{i,j} = \begin{cases} 1_F, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$

Show that $l = \{l_1, l_2, \cdots l_n\}$ is a basis for X'. Such a basis l is called the dual basis of x.

Sol^n

$$l_i : X \to F$$

Since l_i is linear functional.

So, Let $x, y \in X$, $\alpha \in F$

then $l_i(\alpha x + y) = \alpha\, l_i(x) + l_i(y)$

Given $l_i(x_j) = \delta_{ij}$ this map is uniquely determined by its action on basis of X.

Now, T.S.T $l = \{l_1, l_2, \cdots l_n\}$ is a basis for X'

- <u>l spanning X'</u>

Let L be any linear functional in X'.

Since L is linear,

$$L(x_j) = a_j \quad \forall j = 1, 2, \cdots n$$

then, L can be expressed as,

$$L = a_1 l_1 + a_2 l_2 + \cdots + a_n l_n$$

this shows any linear fun. L in X' is linear Combinations of functional $l_1, l_2, \cdots l_n$.
hence l spans X'.

- <u>linear independent</u>

Let $c_1 l_1 + c_2 l_2 + \cdots + c_n l_n = 0$

In particular

$$(c_1 l_1 + c_2 l_2 + \cdots + c_n l_n) x_i = 0$$

$$\Rightarrow c_1 l_1(x_i) + \cdots + c_i l_i(x_i) + \cdots + c_n l_n(x_i) = 0$$

$$\Rightarrow \quad c_i = 0 \text{ for all } i.$$

hence $l_1, l_2, \cdots l_n$ are L.I.

hence l is a basis of X' which is dual basis.

Q3: Let X be an n-dimensional vector space over F and let 'l' be a non zero linear functional on X.

(i) What is the dim of N_l ?

(ii) If 'f' is a linear functional on X, show that $f = \alpha l$ for some $\alpha \in F$ iff $N_l \subseteq N_f$.

(iii) Let V be a subspace of X with $\dim(V) = n-1$ (such a space is called hyperspace). Find a linear functional $g : X \to F$ s.t. $N_g = V$.

(iv) Find a linear functional 'g' on $\mathbb{R}^2$ such that N_g is the line, $ax + by = 0$, $a, b \in \mathbb{R}$.

Solⁿ. (i) Since X is an n-dim. V.S. over F
& l is a non-zero linear fun. on X.
$l : X \to F$

Now define $N_l = \{x \in X \mid l(x) = 0\}$

Since l is a non-zero
so l maps X onto F

by Rank nullity theoram

$$\dim X = \dim(R_l) + \dim(N_l)$$

$$\Rightarrow n = 1 + \dim(N_l)$$

$$\boxed{\dim N_l = n-1}$$

(ii) T.S.T $f = \alpha l$ for some $\alpha \in F$ iff $N_l \subseteq N_f$

$\Rightarrow$ Assume $f = \alpha l$ for some $\alpha \in F$

$f(x) = \alpha l(x)$ for any $x \in X$.

if $x \in N_l$ then $l(x) = 0$

then $f(x) = \alpha l(x) = \alpha \cdot 0 = 0$

therefore $x \in N_f$

hence $N_l \subseteq N_f$

$\Leftarrow$ Assume $N_l \subseteq N_f$ for any $x \in X$.

then $\exists\, x_0 \in N_f$ s.t. $l(x_0) \neq 0$

Choose $\alpha = \dfrac{f(x_0)}{l(x_0)}$ then,

$f(x) = \alpha l(x)$ for any $x \in X$.

hence $f = \alpha l$ for some $\alpha \in F$.

(iii) Let $g : X \to F$

s.t. $g(v_0) = 1$, $v_0 \in X$

& $g(v) = 0$, $v \in \text{span}(X \setminus \{v_0\})$

Now $N_g = \{v \in X \mid g(v) = 0\}$

$= \text{span}(X \setminus \{v_0\})$

$\dim N_g = n-1$ $(\dim X - \dim(v_0))$.

(Note: we can also take $g = l$ from part (i) which is also has dim(Nullspace) $n-1$)

(iv)

Let Define $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$ s.t.

$$g(x,y) = ax + by \quad , a,b \in \mathbb{R}$$

Clearly g is linear function.

And $$N_g = \{(x,y) \in \mathbb{R}^2 \mid ax+by = 0\}$$

Clearly N_g is the line $ax+by=0$, $a,b \in \mathbb{R}$ $\square$

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